

Unit – 4 : Session – 8 : SLO – 2 | Shared vs Distributed Models — Real-World Application

I. Discussion Question

1. Application: Google Search (large-scale web search engine).
2. Why distributed computing: the web index is petabytes in size — far beyond any single machine — so it is sharded across thousands of servers, each searching its own partition in parallel.
3. Scalability: more machines are simply added as query volume and index size grow (horizontal scaling).
4. Fault tolerance: data and services are replicated, so individual machine failures do not interrupt search results.
5. Low response time: partial results from many nodes are aggregated in parallel, returning answers in milliseconds.
6. Other valid examples: Hadoop/MapReduce batch analytics, Netflix video streaming (CDN), blockchain networks, weather simulation clusters.